

Python

- Getting Started with Python
- Working with Data
- Operators and Expressions
- Control Flow
- Data Structures
- Functions and Advanced Python Concepts
- Practical Applications and Mini-Projects

Getting Started with Python

- Installation
- Print function
- Variables and Data types
- Type casting
- User input

Indentation in Python

- Defines the structure of the code by grouping statements into blocks
- Replaces braces {} used in other languages
- Use 4 spaces per indentation level (Python standard)
- All lines in a block must have the same indentation
- Inconsistent indentation causes errors.

Variables and Data types

A variable is a flexible container for values, and in Python, you can create and modify variables anytime without prior declaration as it is not statically typed.

Data Types

- Strings
- Integers
- Float
- Boolean

Type Casting

The process of converting a variable from one data type to another.

- `str()`
- `int()`
- `float()`
- `bool()`

User Input

- `input()`
 - Prompts the user to enter data
 - Returns the entered data as a string.

Exercises

- Rectangle Area Calculation
- Shopping Cart Program